

# Shivi Gupta

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## EDUCATION

- Carnegie Mellon University** - GPA: 3.92/4.0 Pittsburgh, PA  
Master of Science in Electrical and Computer Engineering May 2025
- Courses: Large Language Models, Computer Vision, Multimodal Machine Learning, CPU and GPU Computing, Parallel Computing, Computer Architecture, Distributed Machine Learning, Optimization, Graph Learning, Ethics in AI
- Indian Institute of Technology Kanpur** - GPA: 9.2/10.0 (Distinction) Kanpur, India  
Bachelor of Technology in Electrical Engineering May 2023
- Courses: Image Processing, Digital Signal Processing, Machine Learning for Signal Processing, Control Systems Analysis

## SKILLS

**Languages:** Python, C++, JavaScript, MATLAB, SQL, OpenMP, OpenGL, CUDA, Verilog, System Verilog  
**Utilities:** PyTorch, Pandas, TensorFlow, NumPy, Scikit-Learn, AWS, Keras, Gem5, Synopsis, Git, Matplotlib, Linux, GCP, Docker, Azure, Kubernetes, Android Studio, Vulkan, System Design

## EXPERIENCE

- Electrical and Computer Engineering, CMU** Pittsburgh, PA  
Research Assistant | Memory Optimization, Neuromorphic Computer Architecture Jan 2025 – May 2025
- Designed a Neuromorphic Content-Addressable Memory (CAM) system using hierarchical reference frames and dynamic memory composition reducing memory footprint by **16x**
  - Built a C++ spatial learning mouse-in-a-maze simulator with sparse, composable data structures for memory constraints
- Schlumberger (SLB)** Cambridge, MA  
Advanced Acoustics Intern | Acoustic Signal Processing, Domain Generalization, Ordinal Regression May 2024 - Aug 2024
- Advanced an acoustic tool by integrating heterogeneities and uncertainty metrics to generate high resolution ultrasonic-like features from lower resolution sonic data
  - Developed domain generalization and ordinal regression pipelines resulting in a **4%** improvement in accuracy
- Inter-University Accelerator Center (IUAC)** Delhi, India  
Research Intern | Intelligent Control Systems, Behavioral Cloning, Expert Policy Learning Jul 2022 - Dec 2022
- Automated control process in a particle accelerator in one of India's first efforts towards intelligent accelerator operation
  - Developed a behavioral cloning model based on expert data for automation and parameter significance estimation
- Adobe** Bangalore, India  
Research Intern | Sequence Modeling, Edge Computing May 2021 - Aug 2021
- Built user profiles using multitask learning and autoencoders, generating **16-dimensional** universal edge-compatible vectors
  - Co-inventor on Adobe **Patent P10960-US** on cloud-to-edge profile migration
- Electrical Engineering, IITK** Kanpur, India  
Undergraduate Researcher | Human-Computer Interaction, Cognitive Modeling, Visual Perception Jan 2021 - Apr 2021
- Created an experiment for data collection on emotion-color links in images, demonstrating a correlation of **0.34** in VGG-16
  - First author on a paper published in the CogSci 2021, 43rd edition "Emotion-Color Association in Biologically Inspired Deep Neural Networks" (Gupta, S., & Gupta, S. 2021)
- ## PROJECTS
- GPU-Accelerated OpenCV GrabCut** | GPU & CPU Acceleration, OpenMP, CUDA Jan 2025 - May 2025
- Accelerated GrabCut segmentation using CUDA to parallelize K-Means and Graph weight computation by optimizing memory access with coalesced loads and shared memory reuse, achieving a **120x** speedup
- Evaluating Persuasiveness of Language Models** | LLMs, NLP, Instruction tuning, Fine-tuning Sep 2024 - Dec 2024
- Instruction tuned open-sourced LLMs on Anthropic Persuasion dataset, achieving **0.3** Cohen's Kappa in zero shot prompts
  - Fine-tuned GPT models to generate persuasive arguments, resulting in nuanced arguments with a perplexity score of 2082
- 3D Object Grounding with Generative Modeling** | Multimodal ML, Knowledge Distillation Jan 2024 - May 2024
- Fine-tuned a compact 3D generative model on the OmniObject3D dataset using knowledge distillation from a large text-to-3D Stable Diffusion model
  - Integrated text and image modalities via cross-modal fusion to improve representation alignment and generalizability
- Conditional Point Cloud Generation** | Computer Vision, Generative AI, Stable Diffusion Sep 2023 - Dec 2023
- Generated 3D point clouds from text and Canny images using ControlNet and a fine-tuned stable diffusion framework
  - Optimized Stable Diffusion using low-rank adaptation (LoRA) for cleaner intermediate 2D images, and used Wonder-3D for mesh-to-point cloud sampling
- Self-Configuring Graph Convolution Networks** | GNNs, Graph Attention Sep 2023 - Dec 2023
- Proposed a novel self-configuration method for GCNs using K-means based node addition and attention-based edge reweighting, improving accuracy by **3%** on semi-supervised classification